

Andrés Jauch

BSc Data Science Student

Basel / Dornach, Switzerland

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EDUCATION

BSc Data Science - FHNW University of Applied Sciences

2024 – Present

Brugg-Windisch, CH

- Applied focus on statistical modeling, machine learning pipelines, relational data architectures, and software engineering.
- Practical coursework centered on algorithmic problem-solving and reproducible data analysis.

Commercial Apprenticeship & Federal Matura (BM1) - Wirtschaftsschule

2020 – 2023

Basel, CH

- Dual corporate and academic track covering financial accounting, commercial law, and operational business workflows.
- Graduated with Swiss Federal VET Diploma (EFZ) alongside academic baccalaureate (BM).

SELECTED PROJECTS

Rhine Swimability Monte Carlo Simulation | Python, NumPy, SciPy, Pandas

GitHub ↗

- Processed 23 years of multi-sensor daily hydro-climatic time series for the Basel Rhine station.
- Estimated parameters via Maximum Likelihood (MLE) and executed 10'000-iteration bivariate normal simulations with Cholesky decomposition ($\rho \approx 0.50 - 0.57$).
- Implemented a continuous rolling-window evaluation (± 20 days) to prevent discrete month-boundary artifacts.

Tennis 3D Tracking & Analysis | Python, Computer Vision, Stereo Camera, ZED SDK, Streamlit

GitHub ↗

- Built an end-to-end stereo computer vision pipeline computing 3D spatial trajectories for high-velocity ball tracking.
- Implemented court bounce detection, spatial coordinate transformation, and real-time velocity metrics.
- Deployed an interactive Streamlit dashboard visualizing trajectory curves and positional heatmaps.

Brain Health Challenge | Python, Scikit-Learn, Random Forest, Feature Engineering

GitHub ↗

- Engineered feature extraction pipelines from high-dimensional clinical indicator datasets.
- Trained Random Forest classification baselines to predict progression and regression of Alzheimer's Disease.
- Assessed classification thresholds and feature importance rankings using cross-validation and ROC-AUC evaluation.

EXPERIENCE

Legal & Operations Assistant - Advokatur & Rechtsberatung TRIAS AG

2023 – 2024

Basel, CH

- Structured and digitized physical case files into centralized, searchable document archives.
- Automated routine clerical workflows and coordinated deadline-sensitive legal documentation.
- Monitored statutory court deadlines and handled formal correspondence with authorities and clients.

TECHNICAL SKILLS

Languages:

Databases & Data Handling:

Data Science & ML:

Platforms & Tools:

Spoken Languages:

Python, SQL, R, TypeScript, Bash

PostgreSQL, Relational Querying (Joins, Aggregations), Data Cleaning (Pandas)

NumPy, SciPy, Pandas, Scikit-Learn, Matplotlib, Seaborn, Streamlit

Git, GitHub, Docker, Linux

German (Native), Spanish (Native), English (Fluent)